
EXPERIENCE

- Athlete Den** New York, USA
Software Engineer May 2024 – Current
 - Developing patent-pending solution for sports performance analysis; Implemented on-device ML with Python, CNN and CoreML for real-time activity detection using computer vision, LLM, agentic development with 85% model improvement.
 - Developed websites with Next.js and with JS, HTML/CSS by customizing CMS - Wordpress
 - Architected cloud infrastructure on Google Cloud Platform with Firebase for real-time data synchronization, logging and analytics; designed caching for offline use and background sync for visualization.
 - Integrated Label Studio API for data labeling/model training workflows; managed release pipeline with GitHub Actions CI/CD, TestFlight publishing, and Firebase Analytics/crash reporting for easily audit reporting and tracing data.
- New Jersey Institute of Technology + Tennessee Tech University** New Jersey + Tennessee, USA
Graduate Research and Teaching Assistant Sep 2022 – May 2024
 - Developed automated PHI/PII leakage detection systems for Android apps; implemented STRIDE threat modeling aligned with HIPAA-driven data protection needs. Found 61% (38% PHI) of our medical and health applications have at least one leak.
 - Built Python analysis system processing SMS messages using BERT embeddings and BERTopic clustering; achieved 92% detection accuracy and created interactive threat-intelligence visualizations with D3.js dashboards.
 - Led labs for 100+ students/semester; designed programming assignments and hands-on instruction in computer science courses.
- Samsung Research** South Korea & Bangladesh
Lead Software Engineer Jul 2018 – Aug 2022
 - Led development of enterprise-scale Samsung Family Hub application (Backend Server) with secure layered API communication for cloud data sync and robust JSON/XML handling.
 - Engineered multi-threaded local data systems with databases for offline interaction and data integrity; reduced concurrency-related crashes by 99%. Interacted with customers with feedback to provide best error free solution.
 - Architected multi-platform apps (iOS, Android, Watch & Windows) for Samsung Galaxy Buds with BLE serialization/byte-stream handling, authenticated REST APIs, and firmware update system (FOTA).
 - Built automated testing framework (WinAppDriver) and Continuous Integration workflows with vulnerability/I/O anomaly logging and crash analytics for proactive issue detection.
 - Prepared invention disclosures and coordinated IP submissions; evaluated security implications, mapped product features to prior-arts and novelty themes, and contributed technical disclosures during patent documentation reviews. Created 120+ article technical knowledge base while coordinating global cross-platform teams.

TECHNICAL SKILLS

- Language & Systems:** Python, Swift, C#, Flutter, Core ML, Bluetooth protocols, FOTA (Firmware Over-The-Air)
- AI/ML:** Claude Code/Codex AI Assisted Development and Orchestration, Custom LLM/VLM Agent Development, PyTorch, scikit-learn, Neural Networks
- Data Tracing and Visualization:** PowerBI, pandas, NumPy, BigQuery, MySQL, PostgreSQL
- Cloud & Backend:** Google Cloud Platform (Cloud Storage/Compute Engine/Cloud Run), AWS - Knowledge, Firebase (NoSQL), REST APIs, JSON/XML
- DevOps & Testing:** GitHub Actions CI/CD, XCTest, WinAppDriver, TestFlight, Firebase Analytics, Git, Docker

EDUCATION

- New Jersey Institute of Technology + Tennessee Technological University**
M.S. Computer Science (GPA: 4.00/4.00) 2022 – 2024

PATENTS, PUBLICATIONS & RECOGNITION

- Patents:** Application-pending “System, Apparatus, and Method for Analyzing Sports Performance Using Multi-Media Content.”
- Publications:** “SmishViz: Graph-based Visualization System for Smishing Threats,” ACM CODASPY 2025 (20.9% acceptance) | “AbuseGPT: Abuse of Generative AI ChatBots to Create Smishing Campaigns,” IEEE ISDFS 2024.
- Awards:** Eminence Award 2025, Athlete Den Employee of the Year 2026, Samsung R&D ICON of the Month (May 2019, Nov 2020), Outstanding Collaborator (2021).